STUDENT PERFORMANCE ANALYTICS SYSTEM

Python with AI – Project Documentation

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Course: Python with AI

1. Project Objective

The objective of this project is to develop a Python-based Student Performance Analytics System that analyzes student academic performance using Python, NumPy, and Pandas. It calculates total marks, average marks, grades, pass/fail status, subject-wise performance, and identifies top-performing students.

2. Technologies Used

- Python
- NumPy
- Pandas
- Google Colab
- GitHub

3. Dataset Description

The dataset contains 20 student records. It includes Student ID, Name, Department, Python marks, Mathematics marks, Data Science marks, AI marks, Statistics marks, and Attendance.

4. Implementation

1. Created a CSV dataset containing 20 student records.
2. Loaded the dataset using Pandas.
3. Organized the data using a Pandas DataFrame.
4. Created reusable functions for calculations and processing.
5. Calculated total marks and average marks.
6. Assigned grades using conditional statements.
7. Determined pass/fail status based on marks and attendance.

8. Used NumPy for maximum, minimum, and average calculations.
9. Analyzed subject-wise performance.
10. Identified the top three performers.
11. Saved the processed results in a CSV report.

5. Key Features

- Student data loading and display
- Total and average marks calculation
- Grade assignment
- Pass/fail analysis
- Class performance analysis
- Subject-wise performance analysis
- Top three performer identification
- Attendance analysis
- CSV report generation

6. Grading Rule

AVERAGE MARKS	GRADE
90 AND ABOVE	A+
80 TO BELOW 90	A
70 TO BELOW 80	B
60 TO BELOW 70	C
50 TO BELOW 60	D
BELOW 50	F

7. Pass/Fail Criteria

A student passes if every subject mark is at least 40 and attendance is at least 75%. Otherwise, the student is marked as FAIL.

8. Output / Screenshots

```
!python main.py

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 STUDENT PERFORMANCE ANALYTICS SYSTEM
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--- DATASET INFORMATION ---
Total Students: 20
Department: AI&DS

Student Records:
Student_ID          Name Department  Python  Mathematics  Data_Science  AI  Statistics  Attendance
      S001  Aarav Sharma      AI&DS      85           78            88  90          82          92
      S002   Sneha Patil      AI&DS      72           80            75  78          74          88
      S003    Rohan More      AI&DS      65           70            68  72          66          81
      S004  Pranav Joshi      AI&DS      90           92            94  91          89          96
      S005   Ananya Shah      AI&DS      78           85            80  82          79          90
      S006   Om Kulkarni      AI&DS      55           62            58  60          57          76
      S007 Isha Deshmukh      AI&DS      88           86            91  89          87          94
      S008  Aditya Patil      AI&DS      67           73            70  65          69          84
      S009   Neha Jadhav      AI&DS      92           89            95  93          90          97
      S010    Riya Pawar      AI&DS      76           71            79  75          77          87
      S011    Akash Wagh      AI&DS      48           55            52  50          46          68
      S012    Pooja More      AI&DS      83           81            85  87          84          91
      S013  Rahul Shinde      AI&DS      69           64            72  68          70          79
      S014   Kavya Joshi      AI&DS      95           93            96  94          92          98
      S015   Yash Thakur      AI&DS      61           58            65  63          60          73
      S016   Meera Patil      AI&DS      87           84            89  86          88          93
      S017 Vedant Chavan      AI&DS      74           69            76  71          73          85
```

```
!python main.py

--- STUDENT PERFORMANCE ---
Student_ID          Name  Total_Marks  Average_Marks Grade Result
      S001  Aarav Sharma          423           84.6     A   PASS
      S002   Sneha Patil          379           75.8     B   PASS
      S003    Rohan More          341           68.2     C   PASS
      S004  Pranav Joshi          456           91.2    A+   PASS
      S005   Ananya Shah          404           80.8     A   PASS
      S006   Om Kulkarni          292           58.4     D   PASS
      S007 Isha Deshmukh          441           88.2     A   PASS
      S008  Aditya Patil          344           68.8     C   PASS
      S009   Neha Jadhav          459           91.8    A+   PASS
      S010    Riya Pawar          378           75.6     B   PASS
      S011    Akash Wagh          251           50.2     D   FAIL
      S012    Pooja More          420           84.0     A   PASS
      S013  Rahul Shinde          343           68.6     C   PASS
      S014   Kavya Joshi          470           94.0    A+   PASS
      S015   Yash Thakur          307           61.4     C   FAIL
      S016   Meera Patil          434           86.8     A   PASS
      S017 Vedant Chavan          363           72.6     B   PASS
      S018   Sakshi More          405           81.0     A   PASS
      S019   Harsh Borse          261           52.2     D   FAIL
      S020   Diya Mahale          445           89.0     A   PASS

--- CLASS PERFORMANCE ---
```

```
--- CLASS PERFORMANCE ---
Highest Average: 94.00
Lowest Average: 50.20
Class Average: 76.16

--- PASS / FAIL STATISTICS ---
Passed Students: 17
Failed Students: 3

--- SUBJECT-WISE PERFORMANCE ---
Python: 75.35
Mathematics: 75.45
Data_Science: 77.90
AI: 76.75
Statistics: 75.35

Best Subject: Data_Science
Lowest Subject: Python

--- TOP THREE PERFORMERS ---
Kavya Joshi | Average: 94.00 | Grade: A+
Neha Jadhav | Average: 91.80 | Grade: A+
Pranav Joshi | Average: 91.20 | Grade: A+

--- NUMPY ANALYSIS ---
Highest Mark: 96
Lowest Mark: 46
```

```
--- NUMPY ANALYSIS ---
Highest Mark: 96
Lowest Mark: 46
Overall Marks Average: 76.16

--- ATTENDANCE ANALYSIS ---
Average Attendance: 86.30%

Report saved: student_performance_report.csv

ANALYSIS COMPLETED SUCCESSFULLY!
```

9. Final Outcome

The project successfully demonstrates the use of Python, NumPy, and Pandas for student performance analysis. It generates meaningful information about individual students, class performance, subject averages, grades, attendance, and top performers.

10. Challenges and Learning

This project helped in understanding Python functions, loops, conditional statements, CSV file handling, Pandas DataFrames, NumPy calculations, and data analysis. It also provided practical experience in organizing and processing student data.

11. GitHub Repository

Repository URL: https://github.com/praniti14/Student-Performance-Analytics

12. Project Files

- main.py — Main program
- functions.py — Reusable functions
- students.csv — Student dataset
- student_performance_report.csv — Processed report
- README.md — Project overview
- Project_Documentation.docx — Project documentation